Substitution into formulas

1 The perimeter of a triangle is defined by the formula $P = x + y + z$. Find the value of P if the length of each of its three sides are:

a $x = 17\ \text{cm}, y = 16\ \text{cm}$ and $z = 14\ \text{cm}$

b $x = 6\ \text{cm}, y = 3\ \text{cm}$ and $z = 7\ \text{cm}$

2 The perimeter of a square with side lengths of a is given by the formula $P = 4a$. Find P if the length of each side is $5\ \text{cm}$.

3 The perimeter of a rectangle is given by the formula $P = 2 \times (l + w)$, where l is the length and w is the width. If the width of a rectangle is $10\ \text{cm}$ and its length is $5\ \text{cm}$, find its perimeter.

4 The area, A, of triangle is given by the formula $A = \dfrac{bh}{2}$, where h is the height of the triangle and b is the length of its base. For each triangle described below, find the area:

a $b = 3\ \text{cm}$ and $h = 10\ \text{cm}$

b $b = 7\ \text{cm}$ and $h = 5\ \text{cm}$

c $b = 25\ \text{cm}$ and $h = 16\ \text{cm}$

d $b = 12\ \text{m}$ and $h = 0.5\ \text{m}$

5 The area of a square with side lengths of s is given by the formula $A = s^2$. Find the value of A if the length of each side is $6\ \text{cm}$.

6 The area of a rectangle is given by the formula $A = \text{Length} \times \text{Width}$. If the length of a rectangle is $2\ \text{cm}$ and its width is $3\ \text{cm}$, find its area.

7 The area of a rhombus is given by the formula $A = \dfrac{1}{2}xy$, where x and y are the lengths of the diagonals. If the diagonals of a rhombus have lengths of $2\ \text{cm}$ and $4\ \text{cm}$, find the area of the rhombus.

8 For many 3 dimensional shapes, we can find the number of edges, E, on the shape by using the formula $E = V + F - 2$, where V is the number of vertices and F is the number of faces. Find the number of edges of a 3 dimensional shape which has:

a 7 vertices and 7 faces

b 8 vertices and 6 faces

9 The surface area of a rectangular prism is given by formula $S = 2\,(lw + wh + lh)$, where l, w and h are the dimensions of the prism. Given that a rectangular prism has a length of $8\ \text{cm}$, a width of $7\ \text{cm}$ and a height of $9\ \text{cm}$, find its surface area.

10 The volume of a rectangular prism is given by the formula $V = l \times w \times h$, where l , w and h are the dimensions of the prism. Given that a rectangular prism has a length of $4\ \text{cm}$, a width of $8\ \text{cm}$ and a height of $5\ \text{cm}$, find its volume.

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11 The volume of a sphere can be calculated using the formula $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Given that a sphere has a radius of $2\ \text{cm}$, calculate its volume correct to two decimal places.

12 The equation of a straight line is given by the formula $y = mx + c$. Given that $m = 6, x = -4$ and $c = 9$, find the value of y.

13 The simple interest generated by an investment is given by the formula $I = \frac{P \times R \times T}{100}$. Given that $P = \$1000$, $R = 6$ and $T = 7$, find the interest generated.

14 Converting a measure of temperature from Celsius to Fahrenheit is given by the formula:

$$F = 32 + \frac{9C}{5}$$

If $C = 15$, find the value of F.

15 Converting a measure of temperature from Fahrenheit to Celsius is given by the formula:

$$C = \frac{5}{9}(F - 32)$$

If $F = 50$, find the value of C.

16 When the heating system in a house is on a setting of s, the temperature, T, of the house within the first 30 minutes can be estimated by using the formula:

$$T = a + \frac{st}{10}$$

Where a is the initial temperature and t is the number of minutes since turning the heating system on. Calculate the temperature of the room after 18 minutes if the initial temperature is -3°C and the setting on the heater is 5.

17 Dave and Maria are contestants on a game show which has three rounds. Their final scores are based on their performance in each round and are calculated using the formula:

$$\text{Score } = a - b + 5(c + 6)$$

Where a, b and c are their point scores for the first, second and third rounds. The scores are recorded in the following table:

	Round 1	Round 2	Round 3
Dave	-12	0	-5
Maria	-12	18	1

a Calculate Dave's final score.

b Calculate Maria's final score.

c In this game show, the winner is the contestant who has the lowest final score. Which contestant won this game?

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18 Tobias operates his air conditioning unit using a remote with three buttons. The first one raises the temperature by 4°, the second decreases it by 7° and the third increases it by 3°. The air conditioner always starts at 21°.
The temperature setting can be modelled using the equation:

$$T = 4u - 7v + 3w + 21$$

Where u, v and w represent the number of times Tobias presses the first, second and third buttons. Find the temperature setting if Tobias presses the first button 11 times, the second 14 times and the third 10 times.

19 Energy can be measured in many forms. A quantity of energy is given in units of Joules (J). The kinetic energy, E, of an object in motion is calculated using the following formula:

$$E = \frac{mv^2}{2}$$

Where m is the mass of the object in kilograms and v is the speed of the object in metres per second. Find the kinetic energy, E, of an object with a mass of $6\ \mathrm{kg}$, travelling at a speed of $19\ \mathrm{m/s}$.

20 In physics, Newton's second law of motion can be stated as:

$$F = ma$$

Where F is the force acting on an object (in Newtons), m is the mass of the object (in kilograms) and a is the acceleration of that object (in $\mathrm{m/s^2}$).

a Calculate the force imparted by an object with a mass of $660\ \mathrm{g}$ and an acceleration of $13\ \mathrm{m/s^2}$.

b Calculate the acceleration of an object with a mass of $25\ \mathrm{kg}$ and a force of 775 Newtons acting on it.

21 Valerie stands at the top of a cliff and launches a tennis ball across the valley. To estimate the vertical position, y metres, of the ball compared to herself she uses the formula:

$$y = 14.7t - \frac{9.8}{2}t^2$$

Where v is the initial vertical velocity and t is the number of seconds since the ball is launched.

a Find the vertical position of the ball after:

i 2 seconds **ii** 3 seconds **iii** 8 seconds

b At which of the above times is the ball above Valerie?

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